

Splunk Threat Hunting Use Case

1. Description of the TTP:

Boot or Logon AutoStart Execution: Registry Run Keys / Startup Folder:

T1547.001 is a persistence technique where attackers modify Windows Run registry keys or place executables in the Startup folder. This ensures their malicious program automatically runs every time the system starts.

2. Attack Scenario:

- An attacker gains initial access to a system (via malware, phishing, or exploitation).
- To maintain persistence, the attacker adds a malicious entry to `HKCU\Software\Microsoft\Windows\CurrentVersion\Run` or `HKLM\Software\Microsoft\Windows\CurrentVersion\Run`.
- This causes the malware to automatically run every time the user logs in or the system reboots.

3. Required Data Sources:

- **Windows Event Logs:**
 - **Event ID 4657** → Registry value modified.
 - A registry value was changed on the computer so that attacker add malware and make it run automatically at startup.
- **Sysmon Logs:**
 - **Event ID 13** → Registry value set
 - Attackers changes registry value, this makes Windows run the malware every time the computer starts.
 - **Event ID 12 or 14** → Registry create/delete.
 - Attackers may create new Run keys to store persistence.
 - Attackers sometimes remove normal startup entries and replace them with their own.

4. Splunk SPL Search:

- Using Sysmon (**Event Code 12 and 13 and 14**) and Using Windows Security Logs (**Event ID 4657**):

```
(
    index=sysmon (EventCode=12 OR EventCode=13 OR
    EventCode=14)
    TargetObject="*\\Software\\Microsoft\\Windows\\Curr
    entVersion\\Run*" | eval Source="Sysmon" | table
    _time, Source, Computer, User, Image, EventCode,
    TargetObject, Details
)
UNION
(
    index=windows EventCode=4657
    ObjectName="*\\Software\\Microsoft\\Windows\\Curren
    tVersion\\Run*" | eval Source="WindowsSecurity" |
    table _time, Source, ComputerName, SubjectUserName,
    ProcessName, OperationType, ObjectName, NewValue
)
| sort - _time
```

- `_time` → When the event happened.
- `EventCode` → 12 or 13 (type of registry event).
- `Computer` → Which host the event happened on.
- `User` → User who made the change.
- `Image` → The program that changed the registry (e.g., powershell.exe).
- `TargetObject` → Which registry key/folder.
- `Details` → More info about the change.
- `ComputerName` → Which computer.
- `SubjectUserName` → Who did it.
- `ObjectName` → Which registry key.
- `OperationType` → added / modified / deleted.
- `NewValue` → The new value that was written.

5. Detection Logic

- Run registry keys are commonly abused for persistence.
- Sysmon [Event Code 12 and 13 and 14](#) and Security [Event Code 4657](#) record registry value changes.
- By filtering for paths under [HKCU\Software\Microsoft\Windows\CurrentVersion\Run](#) or [HKLM\Software\Microsoft\Windows\CurrentVersion\Run](#), the query finds suspicious modifications that could indicate persistence by malware.

6. Possible False Positives

- Antivirus updates.
- Software auto-update services (Chrome, Teams, etc.).
- User-installed applications adding themselves to startup.

7. Recommended Response and Mitigation

- Verify whether the process being added is legitimate.
- Check the file path of the executable.
- Scan the file with antivirus or sandbox analysis.
- Remove unauthorized Run key entries.
- Investigate how the attacker gained access.
- Reset compromised user credentials.

For more details → <https://attack.mitre.org/techniques/T1547/001/>